LESSON PLAN

AIR WEAPON (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Stability and control: definition, notation used Stability requirements, a/c controls and Maneuvers, static, dynamic stability.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Stability and control: definition, notation used Stability requirements, a/c controls and Maneuvers, static, dynamic stability.
5. Specific Objectives	10.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer and various concepts of Aerodynamics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of aircraft control surface diagrams and maneuver definitions from IB/PDF; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the term 'stability' in the context of aircraft motion.
(b) List the primary flight controls used to maintain longitudinal stability.
(c) Explain the relationship between center of gravity and static stability.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Stability and Control • Notation used for Stability Axes • Stability Requirements for Safe Flight	L & Ds	PPT, IB	15 min
(b)	• Aircraft Controls and Maneuvers • Static Stability Characteristics • Dynamic Stability Behavior	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Static and Dynamic Stability • Role of Damping in Dynamic Response • Summary of Stability Criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Differentiate between static stability and dynamic stability with examples.
(b) Describe how the rudder contributes to directional stability during a turn.
(c) Identify the notation used for the lateral axis in aircraft stability analysis.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from static to dynamic stability as covered in the next lesson on Drag analysis (LP 12).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Drag analysis, induced, form, skin friction, profile parasite and total drag.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min